

Dynamic Guideline: WHO-Based Echinococcosis - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Latest WHO clinical guideline

- **Guideline title:** *WHO guidelines for the treatment of patients with cystic echinococcosis*
 - **Publication date:** 16 June 2025
 - **Disease scope:** **Cystic echinococcosis (CE)** caused by *Echinococcus granulosus* sensu lato; the guideline explicitly does **not** cover alveolar echinococcosis (AE) as its main therapeutic scope.
 - **Target population:** Adults and children with CE cysts.
 - **Primary objective:** To guide treatment choice so patients receive appropriate, equitable, affordable and safe treatment, while avoiding unnecessary invasive procedures and iatrogenic complications where possible.
 - **Target users:** Clinicians, health-facility managers and practitioners at primary, secondary and tertiary care levels; policymakers, health-system administrators, insurers and neglected tropical disease programme implementers.
 - **Core clinical approach:** Imaging-based, cyst-stage and cyst-size-specific management, taking into account symptoms, cyst complications, patient factors, health-care infrastructure and local expertise.
- Sources: [WHO guideline page](#), [WHO IRIS PDF](#), [NCBI Bookshelf](#).

Exactly five WHO / WHO-linked echinococcosis guideline or technical documents synthesized

#	WHO / WHO-linked document	Date	Role in this report
1	<i>WHO guidelines for the treatment of patients with cystic echinococcosis</i>	16 June 2025	Latest WHO clinical guideline; primary basis for CE treatment recommendations. WHO
2	<i>Expert consensus for the diagnosis and treatment of cystic and alveolar echinococcosis in humans</i> — WHO-IWGE / Brunetti et al.	Consensus 2009; published April 2010	WHO-linked expert consensus covering CE and AE diagnosis, treatment and follow-up. WHO publication record
3	<i>WHO/OIE Manual on Echinococcosis in Humans and Animals</i>	Preface dated December 2000	One Health manual addressing human and animal echinococcosis, surveillance, diagnosis, treatment progress and prevention. WHO/OIE Manual PDF
4	<i>WHO Guidelines for Surveillance, Prevention and Control of Echinococcosis/Hydatidosis</i> , second edition	1984	Historical WHO surveillance, prevention and control guidance for humans and animals, referenced in the WHO/OIE manual. WHO/OIE Manual PDF

#	WHO / WHO-linked document	Date	Role in this report
5	WHO fact sheet: <i>Echinococcosis</i>	17 May 2021	Current WHO public-health summary covering disease forms, transmission, prevention, WHO-IWGE classification and WHO activities. WHO fact sheet

WHO guideline-development methodology note: meaningful youth engagement

A WHO evidence brief titled “**Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)**” was identified as a WHO guideline-development case study, not as an echinococcosis guideline. WHO’s launch webinar for this evidence brief was dated **11 December 2025** and described youth engagement as a way to strengthen dialogue, enrich policy development and improve equity in guideline development. No analogous WHO youth-engagement case study for echinococcosis was found. The 2025 CE guideline documentation states that the Guideline Development Group did **not** include patients or patient representatives.

Sources: [WHO webinar page](#), [WHO abortion care guideline page](#), [WHO IRIS PDF](#).

1.2 Key Recommendations

GRADE interpretation used in this report

The 2025 WHO CE guideline used the **GRADE** approach and evidence-to-decision frameworks. WHO Web Annex A contains evidence syntheses, PICO questions, search strategies, GRADE tables, risk-of-bias assessments and ongoing studies; Web Annex B contains evidence-to-decision judgments. The accessible summaries identify the GRADE framework but do not expose every full recommendation statement or all numerical evidence-profile judgments. Therefore, the table below reports:

- 1. WHO-stated certainty or strength where explicitly available; and
- 2. report-applied GRADE ratings where exact WHO annex values were not visible, based on study design, directness, consistency and clinical consensus.

Sources: [WHO guideline page](#), [WHO IRIS PDF](#), [Guideline Central](#).

Recommendation	Applicable condition / population	GRADE Evidence Quality	Strength
Use imaging-based classification to diagnose, stage and guide management of CE; apply a stage-specific approach using cyst stage, size, location, symptoms and complications.	Suspected or confirmed CE, especially hepatic CE.	Moderate	Strong
Manage uncomplicated hepatic CE using cyst stage and size categories, including CE1/CE3a cysts <5 cm, 5–10 cm and >10 cm, and CE2/CE3b cysts ≤5 cm and >5 cm.	Uncomplicated hepatic CE.	Low	Conditional

Recommendation	Applicable condition / population	GRADE Evidence Quality	Strength
Avoid unnecessary invasive treatment for uncomplicated inactive CE cysts; follow with imaging — ultrasound, CT or MRI — using a “watch and wait” approach unless complications occur.	Inactive uncomplicated CE cysts.	Low	Strong
Use surgery as usual treatment for complicated CE liver cysts , especially when complications such as biliary communication, rupture, infection, mass effect or portal hypertension are present.	Complicated hepatic CE.	Low	Strong
For uncomplicated active lung CE cysts ≤5 cm , WHO suggests surgery; do not give albendazole before surgery; give postoperative albendazole if spillage is suspected or occurs. Lung surgery requires high-level surgical capacity.	Uncomplicated pulmonary CE cysts ≤5 cm.	Very Low / expert consensus	Conditional
Consider praziquantel combined with albendazole after percutaneous or surgical procedures for active hepatic CE cyst types CE1, CE2, CE3a and CE3b when clinically indicated to reduce viable protoscolex risk.	Post-procedural hepatic CE.	Very Low	Conditional
Diagnose AE using clinical and epidemiological context plus imaging, histopathology and/or nucleic acid detection, and serology.	Suspected AE.	Low	Strong
For localized AE, pursue early radical, tumour-like surgery when feasible, followed by albendazole prophylaxis.	Resectable AE.	Low	Strong
For unresectable or advanced AE, use continuous albendazole-based therapy as the backbone of care, with individualized interventional measures as needed.	Unresectable or advanced AE.	Low	Strong
Monitor AE patients long term; CDC recommends benzimidazoles continuously for at least 2 years and follow-up for 10 years or more because recurrence can occur.	AE after surgery or under medical therapy.	Low	Strong

Sources: [WHO guideline page](#), [WHO IRIS PDF](#), [WHO expert consensus](#), [WHO health topic page](#), [WHO fact sheet](#), [CDC Clinical Treatment](#).

1.3 Guideline Development Methodology

Evidence review process

The 2025 WHO CE guideline was developed according to the **WHO handbook for guideline development** and international standards for evidence-based guidelines. The process included:

- Prioritization of clinical questions using the **PICO** framework.
- Systematic evidence retrieval and evidence synthesis.
- Extraction of population, intervention, comparator and outcome data.
- Risk-of-bias assessment using the Cochrane Collaboration Risk of Bias tool.
- Pairwise analyses by research question using Cochrane Review Manager.
- Application of the **GRADE** approach to assess certainty of evidence.
- Use of evidence-to-decision frameworks for each PICO question.

Sources: [WHO IRIS PDF](#), [WHO guideline page](#).

Evidence-to-decision criteria

WHO evidence-to-decision judgments considered:

- Desirable and undesirable effects.
- Certainty of evidence.
- Values and preferences.
- Balance of benefits and harms.
- Resource requirements.
- Certainty about resource needs.
- Cost-effectiveness.
- Health equity.
- Acceptability.
- Feasibility.

Source: [WHO IRIS PDF](#).

Consensus method and contributors

The 2025 WHO guideline involved:

- WHO Guideline Steering Group.
- Systematic Review Team.
- Guideline Development Group.
- External Review Group.
- WHO Guideline Review Committee.

The Guideline Development Group included 17 members, representing endemic regions, but did **not** include patients or patient representatives. No youth-specific engagement was documented for the CE guideline.

Source: [WHO IRIS PDF](#).

Conflict-of-interest management

The WHO guideline process used standard WHO guideline-development governance, including declaration and management of interests through WHO's guideline process. The available summaries confirm WHO Guideline Review Committee involvement but do not expose all individual declaration details.

Source: [WHO IRIS PDF](#).

II. Evidence Summary After WHO Latest Guideline Release (WHO guidelines for the treatment of patients with cystic echinococcosis)

(Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search strategy

Evidence was screened for publications and guidance after the WHO CE guideline release on **16 June 2025** through **29 April 2026**, focusing on:

- Cystic echinococcosis diagnosis, staging, treatment, management and outcomes.
- Alveolar echinococcosis diagnosis, treatment, follow-up and prognosis.
- Systematic reviews, meta-analyses, multicenter observational studies, public-health burden analyses and authoritative institutional guidance.
- Clinical trial or registry evidence, where available.

Sources and databases represented in the evidence set

- WHO official publications and technical pages.
- CDC clinical and laboratory guidance.
- NCBI Bookshelf / NIH-hosted WHO guideline copy.
- Peer-reviewed systematic reviews and meta-analyses identified through biomedical indexing and secondary guideline bibliographies.
- CDC *Emerging Infectious Diseases*.
- Guideline summaries and bibliographic sources.

Sources: [WHO guideline page](#), [CDC Clinical Overview](#), [CDC DPDx](#), [CDC Clinical Treatment](#), [Medscape bibliography](#).

Inclusion criteria

Included evidence met at least one of the following:

- Published or date-verified between **16 June 2025** and **29 April 2026**.
- Addressed echinococcosis diagnosis, treatment, management, epidemiologic burden, follow-up or outcomes.
- Included CE, AE or both.
- Was a systematic review, meta-analysis, multicenter observational study, burden analysis or authoritative clinical guidance.

Exclusion criteria

Excluded evidence included:

- Publications clearly before **16 June 2025**, unless needed as background for WHO recommendations.
- Animal-only studies without direct relevance to human clinical or public-health management.
- Non-authoritative narrative sources without identifiable evidence basis.
- Trial-registry items: none were date-verified in the supplied evidence set.

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
Mayer T, Krienke ME, Chastain DB, et al. "A Multicenter Retrospective Study of Cystic and Alveolar Echinococcosis Patients in the United States (1997–2024)." <i>American Journal of Tropical Medicine and Hygiene</i> , 3 February 2026 .	Multicenter retrospective observational study.	Covers CE and AE patients in the United States over 1997–2024; relevant to clinical characterization and management patterns in a non-endemic / mixed-exposure context. Detailed outcome data were not available in the accessible summary. Medscape bibliography	Low
Xiao W, Liu X, Du L, Tian Y, Li C, Ren L. "Comparative analysis of cystic echinococcosis burden trends: a systematic evaluation of global and Chinese regional patterns using Global Burden of Disease Study 2021 data." <i>Parasites & Vectors</i> , 28 December 2025 .	Systematic burden-trend evaluation using GBD 2021 data.	Supports public-health prioritization by evaluating global and Chinese regional CE burden trends. Useful for surveillance, resource allocation and control-program planning rather than direct bedside treatment choice. Medscape bibliography	Low
Can SM, Aldi FI, Sarikaya MB, Serin PS, Sakru N. "Epidemiological, Diagnostic, and Clinical Features of Intracranial Cystic Echinococcosis: A Systematic Review." <i>Pathogens</i> , 10 December 2025 .	Systematic review.	Addresses rare intracranial CE, including epidemiological, diagnostic and clinical features. Reinforces need for imaging-based diagnosis and specialist neurosurgical / infectious disease management for CNS disease. Medscape bibliography	Low
Leonardi R, Curatolo A, Lo Bianco M, et al. "Clinical Presentation, Management and Outcome of Cerebral Echinococcosis in Children: A Systematic Review and Meta-Analysis." <i>Pathogens</i> , 11 November 2025 .	Systematic review and meta-analysis.	Focuses on pediatric cerebral echinococcosis, including clinical presentation, management and outcomes. Highlights the importance of pediatric specialist referral and careful perioperative management. Medscape bibliography	Low
"Alveolar Echinococcosis in the Early 2020s: A Systematic Review."	Systematic review of 248 original-data articles from January 2020–December 2025.	Reports advances in AE diagnosis and management, including imaging, histopathology and molecular diagnostics; identifies persistent problems including delayed diagnosis, misdiagnosis, limited antiparasitic options, lack of human vaccines and need for specialized interdisciplinary AE centers. MDPI , PMC	Low

Study	Design	Key Findings	GRADE Quality
CDC clinical guidance on echinococcosis, current as retrieved.	Authoritative clinical guidance.	For AE, CDC states radical surgery is preferred when suitable; benzimidazoles should be given continuously for at least 2 years; patients should be monitored for 10 years or more; intermittent albendazole is not recommended. CDC Clinical Treatment	Low
CDC DPDx laboratory guidance.	Authoritative laboratory guidance.	CE serologic sensitivity varies around 60–90% depending on case characteristics and antigens; best serologic diagnosis uses combined tests, such as EIA or IHA screening with immunoblot confirmation; cross-reactions can occur, so clinical and epidemiologic context is essential. CDC DPDx	Moderate

2.3 Evidence Gaps and Future Research Needs

Evidence gap	Clinical / public-health implication	Priority research need	GRADE certainty supporting gap
Limited high-quality randomized evidence for CE treatment by cyst stage and size.	Many recommendations rely on observational data, expert consensus and feasibility considerations.	Pragmatic multicenter comparative studies of albendazole, PAIR, catheter-based methods, surgery and watch-and-wait strategies stratified by WHO-IWGE cyst stage.	Low
Incomplete accessible data for full WHO Web Annex GRADE profiles.	Clinicians may find it difficult to compare certainty across treatment options without consulting annexes directly.	Wider dissemination of simplified evidence profiles and decision algorithms.	Low
Rare-site CE, especially cerebral and intracranial disease, is driven largely by case reports and small series.	Management requires specialist referral; general CE recommendations may not apply directly to CNS disease.	International registries and pooled individual-patient-data analyses for CNS CE, especially in children.	Very Low

Evidence gap	Clinical / public-health implication	Priority research need	GRADE certainty supporting gap
AE is frequently diagnosed late and can mimic malignancy.	Delayed diagnosis reduces curative surgical options.	Better diagnostic algorithms integrating imaging, serology, histopathology and molecular tests; evaluation of metagenomic sequencing in difficult cases.	Low
AE therapy remains mainly parasitostatic for unresectable disease.	Long-term therapy and monitoring are needed; cure is difficult without complete resection.	Trials or high-quality cohorts evaluating new antiparasitic regimens, treatment duration, therapeutic drug monitoring and biomarkers of inactive disease.	Low
Limited patient, community and youth engagement in echinococcosis guideline development.	Recommendations may underrepresent lived experience, access barriers and rural endemic-community priorities.	WHO-style participatory guideline-development models adapted for neglected zoonotic diseases.	Low
Surveillance remains fragmented and underreported in many regions.	Burden estimates and resource allocation may be inaccurate.	Integrated One Health surveillance linking human, veterinary, slaughterhouse and environmental data.	Low

Sources: [WHO guideline page](#), [WHO IRIS PDF](#), [WHO fact sheet](#), [PMC AE systematic review](#), [CDC Clinical Treatment](#).

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
Use imaging-based, stage-specific management for CE.	Post-guideline evidence does not overturn this approach; CDC and WHO-IWGE-consistent sources continue to support imaging as central to diagnosis, staging and management.	Maintain. Use ultrasound, CT or MRI to define cyst stage, size, location, complications and treatment response. GRADE: Moderate; Strength: Strong.

Original Recommendation	New Evidence Impact	Updated Recommendation
Manage uncomplicated inactive CE cysts with watch-and-wait imaging follow-up and avoid surgery unless complications occur.	No new evidence identified that justifies routine intervention for inactive uncomplicated cysts.	Maintain. Continue non-invasive follow-up with imaging; intervene only if complications, diagnostic uncertainty or clinically significant mass effect arise. GRADE: Low; Strength: Strong.
Treat complicated CE liver cysts surgically when clinically appropriate.	No new comparative evidence changes this recommendation. Rare-site and complicated disease reviews reinforce need for specialist management.	Maintain. Refer complicated hepatic CE to experienced hepatobiliary / infectious disease teams; combine surgery with antiparasitic planning when indicated. GRADE: Low; Strength: Strong.
For uncomplicated lung CE cysts ≤ 5 cm, WHO suggests surgery, no preoperative albendazole, and postoperative albendazole if spillage is suspected or occurs.	No new date-verified evidence was found that changes this expert-consensus recommendation.	Maintain with caution. Manage in surgical centers with lung CE experience; document spillage risk and postoperative albendazole indication. GRADE: Very Low; Strength: Conditional.
Diagnose CE using imaging supported by serology and epidemiologic context.	CDC DPDx emphasizes variable serologic sensitivity and need for combined testing with confirmatory immunoblot where available.	Clarify. Do not exclude CE solely on negative serology when imaging and exposure history are compatible; use combined serologic testing when possible. GRADE: Moderate for diagnostic principle; Strength: Strong.
Diagnose AE through combined clinical, epidemiologic, imaging, histopathologic/molecular and serologic evidence.	The 2020–2025 AE systematic review supports expanded use of histopathology and molecular methods, including metagenomic sequencing in difficult cases.	Strengthen. In suspected AE, especially malignancy-mimicking liver lesions, pursue multidisciplinary diagnostic confirmation and consider molecular testing when standard methods are inconclusive. GRADE: Low; Strength: Strong.
For localized AE, perform radical surgery when feasible followed by albendazole prophylaxis.	No new evidence challenges this standard; current reviews continue to describe surgery as primary treatment when resection is possible.	Maintain. Refer early to specialized hepatobiliary / AE centers to maximize curative resection opportunity. GRADE: Low; Strength: Strong.

Original Recommendation	New Evidence Impact	Updated Recommendation
For unresectable AE, use continuous albendazole-based therapy and individualized interventions.	CDC guidance confirms continuous benzimidazole therapy for at least 2 years and monitoring for 10 years or more; intermittent albendazole is not recommended.	Maintain and operationalize. Use continuous benzimidazole therapy, monitor toxicity and adherence, and arrange long-term imaging / serologic follow-up. GRADE: Low; Strength: Strong.
Surveillance and control should be One Health-oriented.	New burden analyses support ongoing need for regional prioritization and surveillance strengthening.	Strengthen. Link human case detection with veterinary control, dog deworming, slaughterhouse hygiene, livestock measures, public education and burden mapping. GRADE: Low; Strength: Strong.

Sources: [WHO guideline page](#), [WHO expert consensus](#), [CDC Clinical Overview](#), [CDC DPDx](#), [CDC Clinical Treatment](#), [PMC AE systematic review](#), [Medscape bibliography](#).

Practical Clinical Algorithm

Suspected cystic echinococcosis

1. Assess exposure risk

- Residence in or travel to endemic areas.
- Contact with dogs, sheepdogs or livestock.
- Ingestion risk from contaminated food, water or soil.

2. Perform imaging

- Ultrasound for hepatic cyst screening and staging.
- CT or MRI for complex anatomy, complications, extrahepatic disease or pre-procedure planning.

3. Classify cyst

- Use WHO-IWGE imaging classification.
- Identify active, transitional or inactive cyst morphology.
- Record cyst size, number, location and complications.

4. Use serology as supportive evidence

- Use EIA or IHA screening where available.
 - Confirm positives with immunoblot if possible.
 - Interpret negative serology cautiously because sensitivity varies.
- Source: [CDC DPDx](#).

5. Select management

- Inactive uncomplicated cyst: watch-and-wait imaging follow-up.
- Complicated hepatic cyst: surgical evaluation.
- Active uncomplicated hepatic cyst: stage- and size-specific WHO-guided therapy.
- Pulmonary cyst: surgical evaluation, especially for active cysts.
- CNS, cardiac, vascular or other rare sites: urgent specialist referral.

Suspected alveolar echinococcosis

1. Recognize malignancy-like hepatic disease

- AE commonly involves the liver and may invade locally or spread hematogenously.
Source: [CDC Clinical Overview](#).

2. Confirm diagnosis using combined modalities

- Imaging.
- Serology.
- Histopathology when safe and needed.
- Molecular testing where available.

3. Determine resectability

- Refer to hepatobiliary surgery and infectious disease specialists.
- Radical surgery is preferred when feasible.

4. Treat medically

- Use continuous benzimidazole therapy when indicated.
- Avoid intermittent albendazole for AE.
- Monitor long term, often 10 years or more.
Source: [CDC Clinical Treatment](#).

Overall Conclusion

As of **29 April 2026**, the most recent WHO echinococcosis clinical guideline is the **16 June 2025 WHO guidelines for the treatment of patients with cystic echinococcosis**. It supports imaging-based, stage- and size-specific management of CE, avoidance of unnecessary invasive procedures, watch-and-wait follow-up for inactive uncomplicated cysts, surgery for complicated hepatic disease and conditional expert-consensus guidance for small active pulmonary CE cysts. For AE, current WHO-linked consensus and CDC guidance continue to support early diagnosis, radical surgery when feasible, albendazole prophylaxis or continuous therapy, individualized interventions and long-term follow-up.

Post-guideline evidence through **29 April 2026** does **not** warrant major reversal of WHO recommendations. It reinforces the need for specialist care for rare-site CE, improved AE diagnostic pathways, long-term AE monitoring, better surveillance and higher-quality comparative treatment studies.